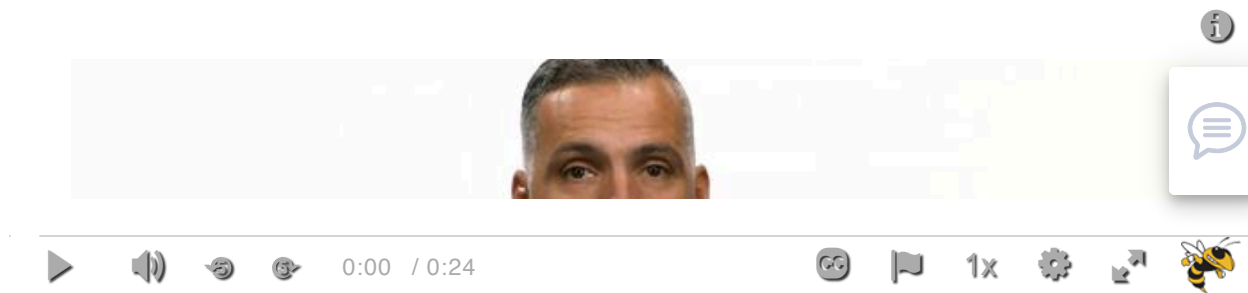


Lesson Two Overview



Throughout the lecture, we are giving you some **“food for thought”** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Review relevant concepts from graph theory and linear algebra
- Review basic graph algorithms
- Explain how these concepts, math, algorithms relate to real-world networks

Required and Recommended Reading

Required Reading

- Chapter-2 from A-L. Barabási, [Network Science](http://www.networksciencebook.com/) (<http://www.networksciencebook.com/>), 2015.
- Chapter-2 from D. Easley and J. Kleinberg, [Networks, Crowds and Markets](http://www.amazon.com/Networks-Crowds-Markets-Reasoning-)  (<http://www.amazon.com/Networks-Crowds-Markets-Reasoning->

[Connected/dp/0521195330](#)), Cambridge Univ Press, 2010 ([Online](#) 
(<https://www.cs.cornell.edu/home/kleinber/networks-book>)_).

Recommended Reading

- *Fibonacci Heap*. Wikipedia. https://en.wikipedia.org/wiki/Fibonacci_heap 
(https://en.wikipedia.org/wiki/Fibonacci_heap)
- *Kosaraju's Algorithm*. Wikipedia.
https://en.wikipedia.org/wiki/Kosaraju%27s_algorithm 
(https://en.wikipedia.org/wiki/Kosaraju%27s_algorithm)

Assignments

 L2: Knowledge Check

L2: Quiz

Module One: Assignment